

Question 1 (1 pt)

Operations, such as finding the height, determining the number of nodes, checking whether the tree is empty, tree traversal, and so on, on AVL trees cannot be implemented the same way they are implemented on binary trees.

- ☐ True
 - ☐ False
-

Question 2 (1 pt)

There are two types of AVL tree rotations: ____.

- ☐ inner and outer
 - ☐ left and right
 - ☐ right rotation and left inversion
 - ☐ inner and left
-

Question 3 (1 pt)

In AVL tree, delete operation involves rotations that may happen up till root.

- ☐ True
 - ☐ False
-

Question 4 (1 pt)

Height difference at any node in AVL must not be.

- ☐ negative
- ☐ greater than 1
- ☐ greater than 2

- ☐ positive
-

Question 5 (1 pt)

When deleting from a 2-4-tree, if the leaf contains only the minimum number of keys, look at the sibling nodes that are adjacent to the leaf.

- ☐ True
 - ☐ False
-

Question 6 (1 pt)

In Red-Black Tree, root must have red children.

- ☐ True
 - ☐ False
-

Question 7 (1 pt)

Black height for every external node is same.

- ☐ True
 - ☐ False
-

Question 8 (1 pt)

Color information can take ____ bit for Red-Black Tree.

- ☐ 1
- ☐ 2
- ☐ 8
- ☐ 4

Question 9 (1 pt)

Insertion in Red-Black Tree will give what kind of problem.

- ☐ black height
 - ☐ double red
 - ☐ both 1 and 2
 - ☐ none
-

Question 10 (1 pt)

In Red-Black Tree, red node can have both black and red children.

- ☐ True
- ☐ False